



## INFLAMMATORY BOWEL DISEASE IBD - L 1

### The AIM of this lecture is

- To know the common diseases of large bowel
- To Know the management of IBD

### ❖ Intended learning outcomes:

#### • By the end of this lecture the student will be able to:

- Deal with common diseases of Large bowel.
- Manage cases of IBD .

ملاحظة : مبدئيا راح تشوفون المحاضرة  
بيها كلام كثير والسبب انه محاضرة الاصلية  
(البوربوينت اللي انشרכת) كانت عبارة عن  
رؤوس نقاط اما الشرح نسخته من المصدر  
وفق متطلبات المحاضرة حتى تكون  
المعلومات كاملة ، والله ولي التوفيق .

## INFLAMMATORY BOWEL DISEASE IBD

- Ulcerative colitis and Crohn's disease are chronic inflammatory bowel diseases which have a protracted relapsing and remitting course, usually extending over years.
- The diseases have many similarities, and it is sometimes impossible to differentiate between them.
- Age : Second and third decades of life, with a second smaller incidence peak in the seventh decade.
- Life expectancy in patients with IBD is like that of the general population.

| 21.51 Comparison of ulcerative colitis and Crohn's disease |                                                                                                                |                                                                                                                                                                                                           |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | Ulcerative colitis                                                                                             | Crohn's disease                                                                                                                                                                                           |
| Age group                                                  | Any                                                                                                            | Any                                                                                                                                                                                                       |
| Gender                                                     | M = F                                                                                                          | Slight female preponderance                                                                                                                                                                               |
| Incidence                                                  | Stable                                                                                                         | Increasing                                                                                                                                                                                                |
| Ethnic group                                               | Any                                                                                                            | Any; more common in Ashkenazi Jews                                                                                                                                                                        |
| Genetic factors                                            | HLA-DR*103; colonic epithelial barrier function (HNF4α, LAMB1, CDH1)                                           | Defective innate immunity and autophagy (NOD2, ATG16L1, IRGM)                                                                                                                                             |
| Risk factors                                               | More common in non-/ex-smokers<br>Appendectomy protects                                                        | More common in smokers                                                                                                                                                                                    |
| Anatomical distribution                                    | Colon only; begins at anorectal margin with variable proximal extension                                        | Any part of gastrointestinal tract; perianal disease common; patchy distribution, skip lesions                                                                                                            |
| Extra-intestinal manifestations                            | Common                                                                                                         | Common                                                                                                                                                                                                    |
| Presentation                                               | Bloody diarrhoea                                                                                               | Variable; pain, diarrhoea, weight loss all common                                                                                                                                                         |
| Histology                                                  | Inflammation limited to mucosa; crypt distortion, cryptitis, crypt abscesses, loss of goblet cells             | Submucosal or transmural inflammation common; deep fissuring ulcers, fistulae; patchy changes; granulomas                                                                                                 |
| Management                                                 | 5-ASA; glucocorticoids; azathioprine; biological therapy (anti-TNF, anti-α4β7 integrin); colectomy is curative | Glucocorticoids; azathioprine; methotrexate; biological therapy (anti-TNF, anti-α4β7 integrin); nutritional therapy; smoking cessation; surgery for complications is not curative; 5-ASA is not effective |

(5-ASA = 5-aminosalicylic acid; TNF = tumour necrosis factor)



## Pathophysiology of IBD

### Genetic

- Both CD and UC common in Ashkenazi Jews
- 10% have first-degree relative/ $\geq 1$  close relative with IBD
- High concordance in identical twins (40–50% CD; 20–25% UC)
- IBD associated with other inflammatory conditions (esp. ankylosing spondylosis and psoriasis)

### Environmental

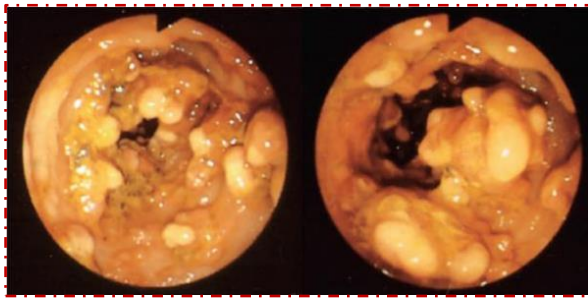
- UC more common in non-smokers and ex-smokers
- CD more common in smokers (relative risk = 3)
- CD associated with low-residue, high-refined-sugar diet
- Commensal gut microbiota altered (dysbiosis) in CD and UC
- Appendicectomy protects against UC

| i                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 21.52 Factors associated with the development of inflammatory bowel disease |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Genetic</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                             |
| <ul style="list-style-type: none"> <li>Both CD and UC are common in Ashkenazi Jews</li> <li>10% have first-degree relative/1 or more close relative with IBD</li> <li>High concordance in identical twins (40–50% CD; 20–25% UC)</li> <li>163 susceptibility loci identified at genome-wide levels of significance; most confer susceptibility to both CD and UC; many are also susceptibility loci for other inflammatory conditions (especially ankylosing spondylosis and psoriasis)</li> <li>UC and CD are both associated with genetic variants at HLA locus, and with multiple genes involved with immune signalling (especially IL-23 and IL-10 pathways)</li> <li>CD is associated with genetic defects in innate immunity and autophagy (<i>NOD2</i>, <i>ATG16L1</i> and <i>IRGM</i> genes)</li> <li>UC is associated with genetic defects in barrier function</li> <li><i>NOD2</i> is associated with ileal and stricturing disease, and hence a need for resectional surgery</li> <li><i>HLA-DR*103</i> is associated with severe UC</li> </ul> |                                                                             |
| <b>Environmental</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                             |
| <ul style="list-style-type: none"> <li>UC is more common in non-smokers and ex-smokers</li> <li>CD is more common in smokers (relative risk = 3)</li> <li>CD is associated with a low-residue, high-refined-sugar diet</li> <li>Commensal gut microbiota are altered (dysbiosis) in CD and UC</li> <li>Appendicectomy protects against UC</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                             |
| (CD = Crohn's disease; HLA = human leucocyte antigen; IBD = inflammatory bowel disease; IL = interleukin; UC = ulcerative colitis)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                             |



### Ulcerative colitis

- Inflammation invariably involves the rectum (proctitis) and spreads proximally in a continuous manner to involve the entire colon in some cases (pancolitis).
- In long-standing pancolitis, the bowel can become shortened and post-inflammatory pseudopolyps' develop; these are normal or hypertrophied residual mucosa within areas of atrophy.
- The inflammatory process is limited to the mucosa and spares the deeper layers of the bowel wall.
- Both acute and chronic inflammatory cells infiltrate the lamina propria and the crypts ('cryptitis').
- Crypt abscesses are typical.



### Crohn's disease

- The sites most commonly involved are, in order of frequency, the terminal ileum and right side of colon, colon alone, terminal ileum alone, ileum and jejunum.
- The entire wall of the bowel is edematous and thickened, and there are deep ulcers which often appear as linear fissures; thus the mucosa between them is described as cobblestone .'
- These may penetrate through the bowel wall to initiate abscesses or fistulae involving the bowel, bladder, uterus, vagina and skin of the perineum.
- Mesenteric lymph nodes are enlarged and the mesentery is thickened.
- Crohn's disease has a patchy distribution and the inflammatory process is interrupted by islands of normal mucosa.
- On histological examination, the bowel wall is thickened with a chronic inflammatory infiltrate throughout all layers.

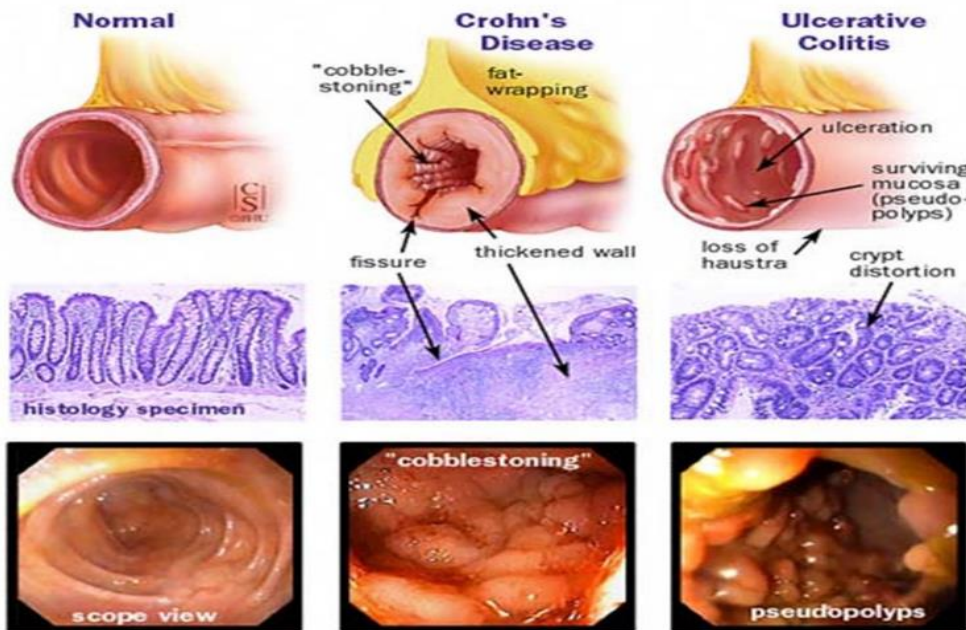


Figure 4. Gross (top), histological (center), and endoscopic (bottom) appearance of normal colon, Crohn's disease, and ulcerative colitis.

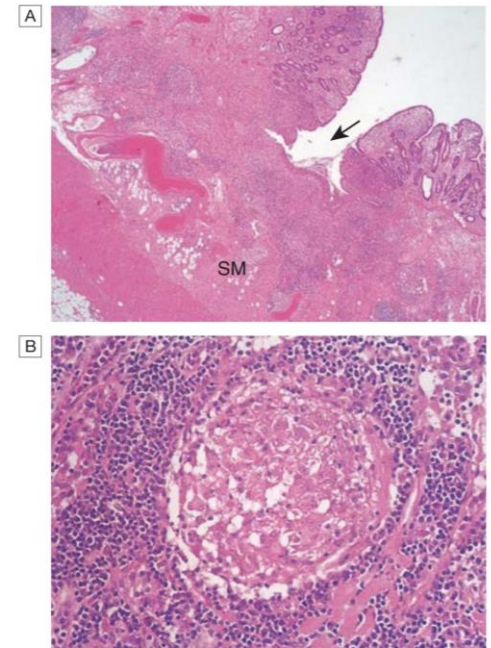


Fig. 21.48 Histology of Crohn's disease. [A] Inflammation is "transmural"; there is fissuring ulceration (arrow), with inflammation extending into the submucosa (SM). [B] At higher power, a characteristic non-caseating granuloma is seen.

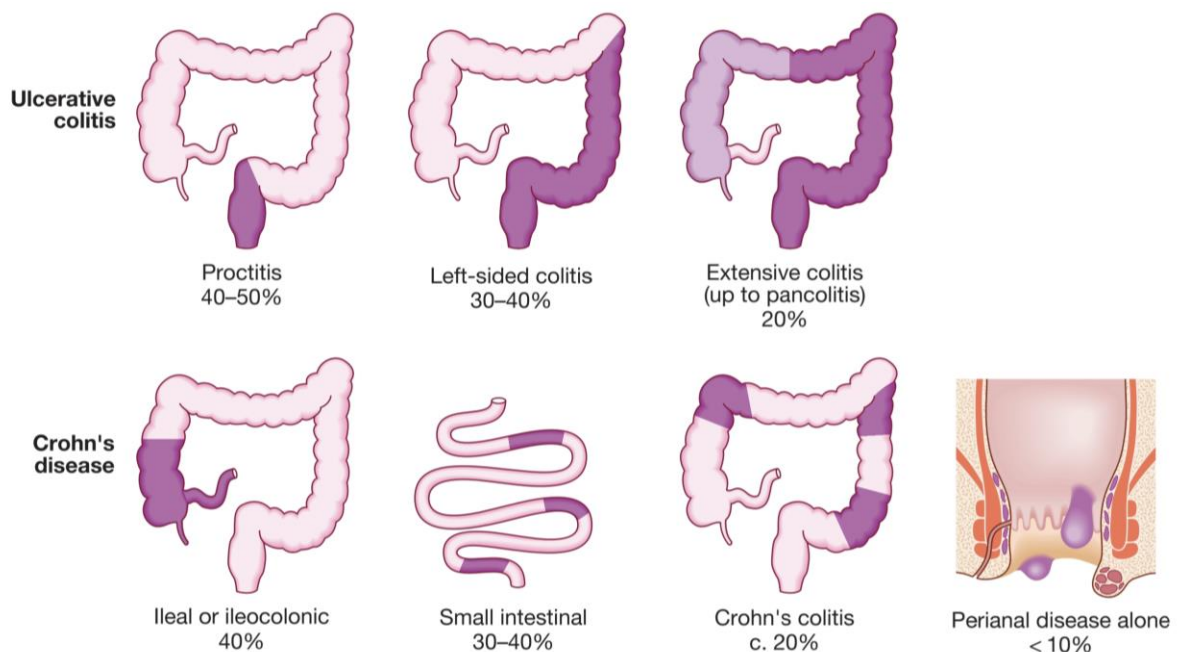


Fig. 21.45 Common patterns of disease distribution in inflammatory bowel disease.



### Clinical features of IBD:

#### Ulcerative colitis

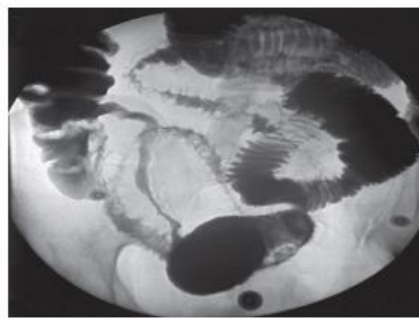
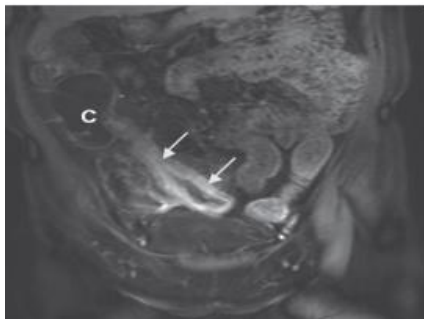
- The cardinal symptoms are rectal bleeding with passage of mucus and bloody diarrhoea.
- The presentation varies, depending on the site and severity of the disease , as well as the presence of extra-intestinal manifestations.
- The first attack is usually the most severe and is followed by relapses and remissions.
- Emotional stress, intercurrent infection, gastroenteritis, antibiotics or NSAID therapy may all provoke a relapse.
- Proctitis causes rectal bleeding and mucus discharge, accompanied by tenesmus.
- Some patients pass frequent, small-volume fluid stools, while others pass pelleted stools due to constipation upstream of the inflamed rectum.
- Constitutional symptoms do not occur.
- Left-sided and extensive colitis causes bloody diarrhoea with mucus, often with abdominal cramps.
- In severe cases, anorexia, malaise, weight loss and abdominal pain occur and the patient is toxic, with fever, tachycardia and signs of peritoneal inflammation

| 21.53 Assessment of disease severity in ulcerative colitis |                                    |                  |                                                                                  |
|------------------------------------------------------------|------------------------------------|------------------|----------------------------------------------------------------------------------|
|                                                            | Mild                               | Moderate         | Severe                                                                           |
| Daily bowel frequency                                      | <4                                 | 4–6              | ≥6*                                                                              |
| Blood in stools                                            | +/-                                | +/++             | +++                                                                              |
| Stool volume                                               | <200 g/24 hrs                      | 200–400 g/24 hrs | >400 g/24 hrs                                                                    |
| Pulse                                                      | <90 beats/min                      | <90 beats/min    | ≥90 beats/min*                                                                   |
| Temperature                                                | Normal                             | Normal           | ≥ 37.8°C*                                                                        |
| Haemoglobin                                                | Normal                             | Normal           | <100 g/L (<10 g/dL)*                                                             |
| Erythrocyte sedimentation rate                             | Normal                             | Normal           | >30 mm/hr* (or equivalent C-reactive protein)                                    |
| Serum albumin                                              | >35 g/L (>3.5 g/dL)                |                  | <30 g/L (<3 g/dL)                                                                |
| Abdominal X-ray                                            | Normal                             | Normal           | Dilated bowel, mucosal islands, thumb-printing of mucosa, or absence of features |
| Sigmoidoscopy                                              | Normal or erythema/granular mucosa |                  | Severe mucosal inflammatory changes; ulceration; blood in lumen                  |

\*The Truelove–Witts criteria for acute severe ulcerative colitis are ≥6 bloody stools/24 hrs plus one or more of: anaemia, fever, tachycardia and high inflammatory markers.

**Crohn's disease**

- The major symptoms are abdominal pain, diarrhoea and weight loss.
- Ileal Crohn's disease may cause subacute or even acute intestinal obstruction.
- The pain is often associated with diarrhoea, which is usually watery and does not contain blood or mucus.
- Almost all patients lose weight because they avoid food, since eating provokes pain.
- Weight loss may also be due to malabsorption and some patients present with features of fat, protein or vitamin deficiencies.
- Crohn's colitis presents in an identical manner to ulcerative colitis but rectal sparing and the presence of perianal disease are features that favour a diagnosis of Crohn's disease.
- Many patients present with symptoms of both small bowel and colonic disease.
- A few patients present with isolated perianal disease, vomiting from jejunal strictures or severe oral ulceration.
- Physical examination often reveals evidence of weight loss, anaemia with glossitis and angular stomatitis.
- There is abdominal tenderness, most marked over the inflamed area.
- An abdominal mass may be palpable and is due to matted loops of thickened bowel or an intra-abdominal abscess.
- Perianal skin tags, fissures or fistulae are found in at least 50% of patients





## Complications of IBD

### Life-threatening colonic inflammation

- This can occur in both ulcerative colitis and Crohn's colitis. In the most extreme cases, the colon dilates (toxic megacolon) and bacterial toxins pass freely across the diseased mucosa into the portal and then systemic circulation.
- This complication arises most commonly during the first attack of colitis .
- An abdominal X-ray should be taken daily because, when the transverse colon is dilated to more than 6 cm, there is a high risk of colonic perforation, although this complication can also occur in the absence of toxic megacolon.
- Severe colonic inflammation with toxic dilatation is a surgical emergency and most often requires colectomy.

### Haemorrhage

- Haemorrhage due to erosion of a major artery is rare but can occur in both conditions.

### Fistulae

- These are specific to Crohn's disease.
- Enteroenteric fistulae can cause diarrhoea and malabsorption due to blind loop syndrome.
- Enterovesical fistulation causes recurrent urinary infections and pneumaturia.
- An enterovaginal fistula causes a faeculent vaginal discharge.
- Fistulation from the bowel may also cause perianal or ischiorectal abscesses, fissures and fistulae.

Cancer

- The risk of dysplasia and cancer increases with the duration and extent of uncontrolled colonic inflammation.
- Thus patients who have long-standing, extensive colitis are at highest risk.
- Oral mesalazine therapy reduces the risk of dysplasia and neoplasia in ulcerative colitis.
- Azathioprine also seems to reduce the risk of colorectal cancer in ulcerative colitis and Crohn's colitis.
- This protective effect probably extends to any medical treatment that results in sustained healing of the colonic mucosa.
- The cumulative risk for dysplasia in ulcerative colitis may be as high as 20% after 30 years but is probably lower for Crohn's colitis.
- The risk is particularly high in patients who have concomitant primary sclerosing cholangitis for unknown reasons.
- Tumours develop in areas of dysplasia and may be multiple.
- Patients with long-standing colitis are therefore entered into surveillance programmes beginning 10 years after diagnosis.
- Targeted biopsies of areas that show abnormalities on staining with indigo carmine or methylene blue increase the chance of detecting dysplasia and this technique (termed pancolonoscopic chromo-endoscopy) has replaced colonoscopy with random biopsies taken every 10 cm in screening for malignancy.
- The procedure allows patients to be stratified into high-, medium- or low-risk groups to determine the interval between surveillance procedures. Family history of colon cancer is also an important factor to consider.
- If high-grade dysplasia is found, panproctocolectomy is usually recommended because of the high risk of colon cancer.



### Extra-intestinal complications

- Extra-intestinal complications are common in IBD and may dominate the clinical picture.
- Some of these occur during relapse of intestinal disease; others appear to be unrelated to intestinal disease activity .

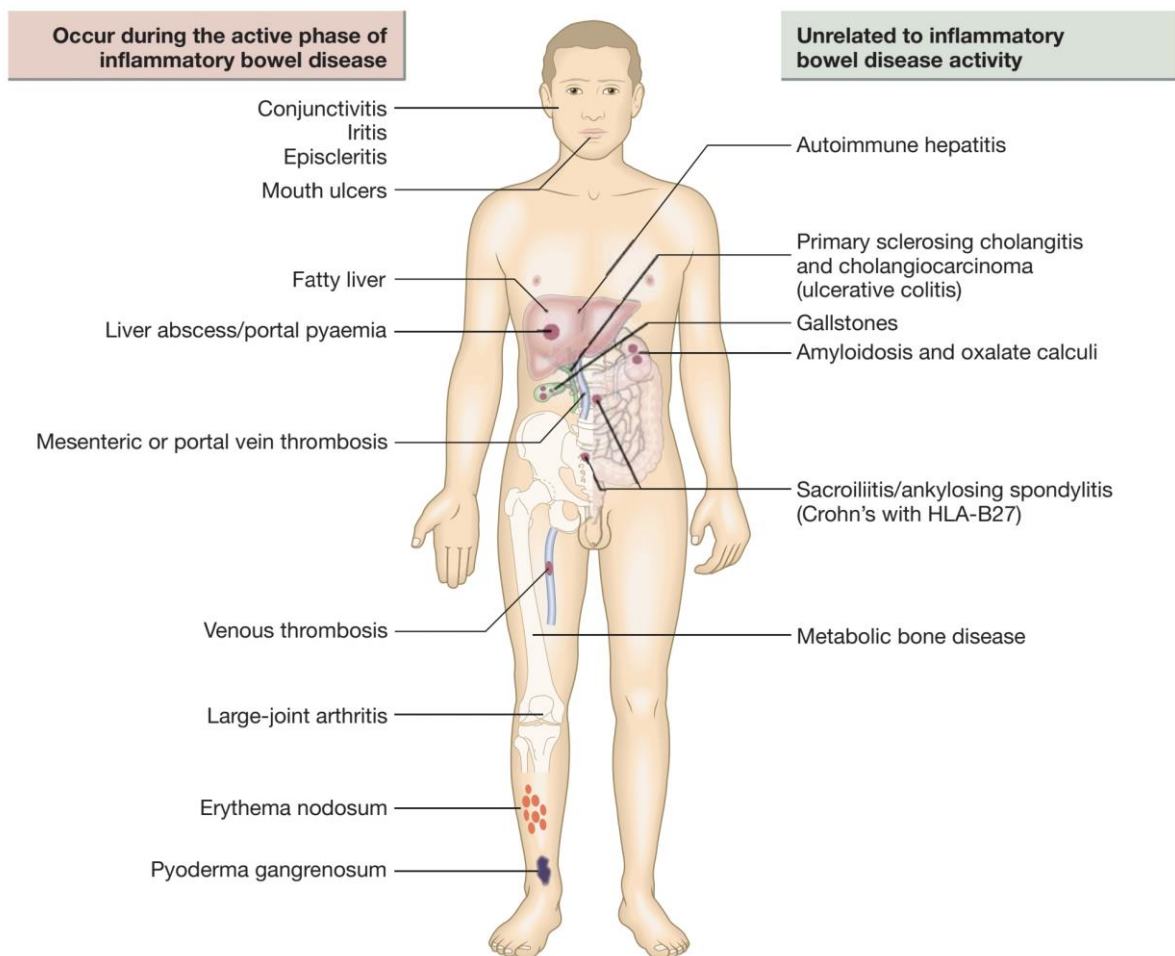


Fig. 21.52 Systemic complications of inflammatory bowel disease. See also Chapters 17 and 18. (HLA = human leukocyte antigen)

### Investigations

- Faecal calprotectin
- GSE
- Endoscopy
- Radiology



## Investigations

- Investigations are necessary to confirm the diagnosis, define disease distribution and activity, and identify complications.
- Full blood count may show anaemia resulting from bleeding or malabsorption of iron, folic acid or vitamin B<sub>12</sub>.
- Platelet count can also be high as a marker of chronic inflammation.
- Serum albumin concentration falls as a consequence of protein-losing enteropathy, inflammatory disease or poor nutrition.
- ESR and CRP are elevated in exacerbations and in response to abscess formation.
- Faecal calprotectin has a high sensitivity for detecting gastrointestinal inflammation and may be elevated, even when the CRP is normal.
- It is particularly useful for distinguishing inflammatory bowel disease from irritable bowel syndrome at diagnosis, and for subsequent monitoring of disease activity.

## Bacteriology

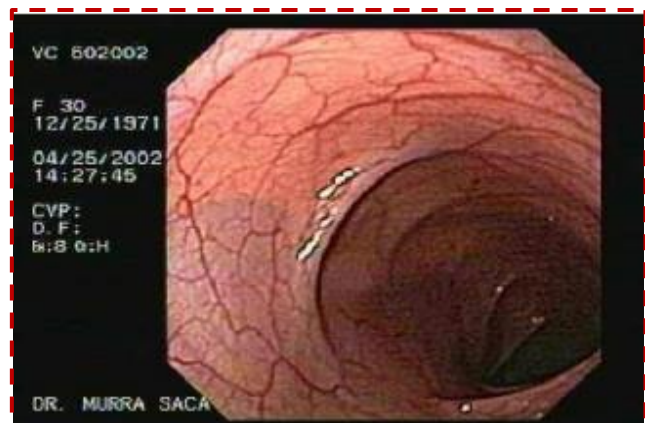
- At initial presentation, stool microscopy, culture and examination for Clostridium difficile toxin or for ova and cysts, blood cultures and serological tests should be performed.
- These investigations should be repeated in established disease to exclude superimposed enteric infection in patients who present with exacerbations of IBD.
- During acute flares necessitating hospital admission, three separate stool samples should be sent for bacteriology to maximise sensitivity.

## Endoscopy

- Patients who present with diarrhoea plus raised inflammatory markers or alarm features, such as weight loss, rectal bleeding and anaemia, should undergo ileocolonoscopy.
- Flexible sigmoidoscopy is occasionally performed to make a diagnosis, especially during acute severe presentations when ileocolonoscopy may confer an unacceptable risk; ileocolonoscopy should still be performed at a later date, however, in order to evaluate disease extent.



- In ulcerative colitis, there is loss of vascular pattern, granularity, friability and contact bleeding, with or without ulceration.
- In Crohn's disease, patchy inflammation, with discrete, deep ulcers, strictures and perianal disease (fissures, fistulae and skin tags), is typically observed, often with rectal sparing.
- In established disease, colonoscopy may show active inflammation with pseudopolyps or a complicating carcinoma.
- Biopsies should be taken from each anatomical segment (terminal ileum, right colon, transverse colon, left colon and rectum) to confirm the diagnosis and define disease extent, and also to seek dysplasia in patients with long-standing colitis guided by pancolonoscopic chromo-endoscopy.
- In Crohn's disease, wireless capsule endoscopy is useful in the identification of small bowel inflammation but should be avoided in the presence of strictures.
- Enteroscopy may be required to make a histological diagnosis of small bowel Crohn's disease, when the inflamed segment is out of reach of standard endoscopes.
- All children and most adults with Crohn's disease should have upper gastrointestinal endoscopy and biopsy to complete their staging.
- Not only is upper gastrointestinal Crohn's disease relatively common in this group, but also it may help to make a definitive diagnosis in patients who otherwise appear to have non-specific colonic inflammation.



Radiology

- Barium enema is a less sensitive investigation than colonoscopy in patients with colitis and, where colonoscopy is incomplete, a CT colonogram is preferred.
- Small bowel imaging is essential to complete staging of Crohn's disease.
- Traditional contrast imaging by barium follow-through demonstrates affected areas of the bowel as narrowed and ulcerated, often with multiple strictures .
- This has largely been replaced now by MRI enterography, which does not involve exposure to radiation and is a sensitive way of detecting extra-intestinal manifestations and of assessing pelvic and perineal involvement.
- These studies use an orally administered small bowel-distending agent and intravenous contrast to provide transmural imaging that can usefully distinguish between predominantly inflammatory strictures (that should respond to anti-inflammatory medical strategies) and fibrotic strictures (that require a mechanical solution, such as surgical resection, stricturoplasty or endoscopic balloon dilatation).
- A plain abdominal X-ray is essential in the management of patients who present with severe active disease.
- Dilatation of the colon, mucosal oedema (thumb-printing) or evidence of perforation may be found.
- Patients with proctitis may have features of proximal faecal loading. In small bowel Crohn's disease, there may be evidence of intestinal obstruction or displacement of bowel loops by a mass.
- Ultrasound is a very powerful tool to detect small bowel inflammation and stricture formation but it is operator-dependent.
- The role of CT is limited to screening for complications, such as perforation or abscess formation, in the acutely unwell.

**Management**

- Drugs that are used in the treatment of IBD are listed in Box 21.56.
- Although medical therapy plays an important role, optimal management depends on establishing a multidisciplinary team- based approach involving physicians, surgeons, radiologists, nurse specialists and dietitians.
- Both ulcerative colitis and Crohn's disease are life-long conditions and have important psychosocial implications; specialist nurses, counsellors and patient support groups have key roles in education, reassurance and coping.
- The key aims of medical therapy are to:
  - + Treat acute attacks (induce remission)
  - + Prevent relapses (maintain remission)
  - + Prevent bowel damage
  - + Detect dysplasia and prevent carcinoma
  - + Select appropriate patients for surgery.



## i

## 21.56 Drugs used in the treatment of inflammatory bowel disease

| Class                                                                                                                 | Mechanism of action                                                                                                                                                                                                                           | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aminosalicylates</b><br>(mesalazine (Asacol, Salofalk, Pentasa, Mezavant), olsalazine, sulfasalazine, balsalazide) | Modulate cytokine release from mucosa<br>Different means of delivery to colon:<br>pH-dependent (Asacol, Salofalk)<br>time-dependent (Pentasa)<br>bacterial breakdown by colonic bacteria from a carrier molecule (sulfasalazine, balsalazide) | No proven value in CD<br>Available as oral or topical (enema/suppository)<br>Sulfasalazine causes side-effects in 10–45%: headache, nausea, diarrhoea, blood dyscrasias<br>Other aminosalicylates better tolerated; diarrhoea, headache in 2–5%<br>Rarely, renal impairment (check urea and electrolytes 6-monthly)                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Glucocorticoids</b><br>(prednisolone, hydrocortisone, budesonide)                                                  | Anti-inflammatory<br>Budesonide is a potent glucocorticoid efficiently cleared from circulation by liver, thereby minimising adrenocortical suppression and steroid side-effects                                                              | Topical, oral or IV, according to disease severity<br>Budesonide considered for active ileitis and ileocolitis<br>High vigilance for complications<br>Never used for maintenance therapy<br>Calcium/vitamin D supplements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Thiopurines</b> (azathioprine, mercaptopurine)                                                                     | Immunomodulation by inducing T-cell apoptosis<br>Azathioprine is metabolised in liver to mercaptopurine, then by TPMT to thioguanine nucleotides                                                                                              | Effective 12 weeks after starting therapy<br>Complications leading to drug withdrawal in approximately 20%: influenza-like syndrome with myalgia, nausea and vomiting; leucopenia in 3%, particularly in inherited TPMT deficiency; hepatotoxicity; pancreatitis<br>60% of those intolerant of azathioprine will tolerate mercaptopurine<br>Increase in lymphoma (approximately 2–3-fold) and non-melanoma skin cancer (life-long sun protection advised)<br>Check TPMT levels prior to starting treatment and avoid if deficient/very low due to risk of toxicity<br>Metabolite levels can be measured to tailor therapy<br>Use with caution for patients presenting over the age of 60 years due to risk of malignancy |
| <b>Methotrexate</b>                                                                                                   | Anti-inflammatory                                                                                                                                                                                                                             | Intolerance in 10–18%<br>Maximal efficacy when given by SC injection once weekly<br>Nausea, stomatitis, diarrhoea, hepatotoxicity and pneumonitis<br>Co-prescription of folic acid and antiemetics.<br>Teratogenic; robust contraception required for males and females                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Ciclosporin</b>                                                                                                    | Inhibits T-cell activation                                                                                                                                                                                                                    | Rescue therapy to prevent surgery in UC responding poorly to glucocorticoids. No value in CD<br>Major side-effects in 0–17%: nephrotoxicity, infections, neurotoxicity (including fits)<br>Minor complications in up to 50%: tremor, paraesthesiae, abnormal liver function tests, hirsutism                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Anti-TNF antibodies</b><br>(infliximab and adalimumab)                                                             | Suppress inflammation and induce apoptosis of inflammatory cells                                                                                                                                                                              | Moderate to severe CD, including fistulating disease<br>Moderate to severe UC and acute severe UC as rescue therapy<br>Acute (anaphylactic) and delayed (serum sickness) infusion reactions after multiple infusions; anti-drug antibody titres and drug levels can be measured<br>Contraindicated in infection; reactivation of latent tuberculosis and moderate to severe cardiac failure<br>Increased risk of infections and possibly of malignancy<br>Rarely, neurological adverse events<br>Requires assessment for latent tuberculosis and hepatitis B and C prior to commencement<br>Continue until treatment failure or 12 months and reassess                                                                   |
| <b>Anti-<math>\alpha 4\beta 7</math> integrin</b><br>(vedolizumab)                                                    | Blocks integrin expressed on leukocytes and inhibits interaction with gut-specific receptor on endothelium, reducing leukocyte migration to gut mucosa                                                                                        | Moderate to severe CD or moderate to severe UC where treatment with anti-TNF has failed or is not tolerated<br>Side-effects include nasopharyngitis, arthralgia, headache<br>Progressive multifocal leukoencephalopathy risk is reduced due to gut specificity<br>Induction with 300 mg infusion at weeks 0, 2 and 6; maintenance 8-weekly infusions thereafter<br>Discontinue if no improvement after 14 weeks<br>Continue until treatment failure or 12 months and reassess                                                                                                                                                                                                                                            |
| <b>Antibiotics</b>                                                                                                    | Antibacterial                                                                                                                                                                                                                                 | Useful in perianal CD and pouchitis<br>Major concern is peripheral neuropathy with long-term metronidazole                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Antidiarrhoeal agents</b><br>(loperamide, co-phenoxylate)                                                          | Reduce gut motility and small bowel secretion<br>Loperamide improves anal function                                                                                                                                                            | Avoided in acute flare-ups of disease<br>May precipitate colonic dilatation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

(CD = Crohn's disease; IV = intravenous; SC = subcutaneous; TNF = tumour necrosis factor; TPMT = thiopurine methyltransferase; UC = ulcerative colitis)